

Top Quantum-Attack Scenarios and Protective Measures (2025)

Quantum computers threaten several cryptographic primitives. The table below summarises ten important quantum-attack “tests” (attack scenarios), the resources/time required according to recent research, a qualitative **risk score** (1 = theoretical/impractical; 10 = imminently feasible), and recommended protective measures.

How to interpret the scores:

- **Feasibility & urgency:** a higher score means fewer qubits/time or broader applicability, so the attack is more likely to be practical soon. For example, harvest-now/decrypt-later attacks score highly because data encrypted today may be stored and decrypted once quantum computers mature. BHT collisions score low because they require impossible amounts of memory.
- **Metrics:** we cite the best public estimates for qubits/time where available; if only complexity is known, we describe it in the “Metrics” column.
- **Protection:** recommended mitigations come from standards bodies (NSA, NIST, IETF) or peer-reviewed research.

#	Attack/test (quantum algorithm)	Metrics & complexity	Risk score	Protection/ mitigation
1	Shor's algorithm on RSA-2048 factoring	Shor's algorithm can factor RSA efficiently. The IETF notes that breaking RSA-2048 on a fault-tolerant quantum computer would require roughly 20 million noisy qubits and ~8 hours of runtime ¹. A 2025 update by Craig Gidney shows that improved techniques (approximate residue arithmetic, magic-state cultivation) reduce the requirement to <1 million noisy qubits and can factor a 2048-bit RSA key in <1 week ² ³. More speculative resource estimates suggest ~600 k qubits could break RSA in ~6–10 hours ⁴.	7/10 – high risk: RSA is ubiquitous; improved algorithms continuously reduce qubit requirements. Breakthroughs like Gidney's 2025 work accelerate the threat timeline.	Migrate to post-quantum cryptography. The IETF warns that no key-size increase will save RSA ⁵. Organisations should adopt lattice- or code-based key-encapsulation mechanisms (e.g., ML-KEM/Kyber, HQC), increase RSA keys to 3072/4096 bits during transition, and apply hybrid key-exchange combining RSA with PQC. Limit the lifetime of RSA keys to reduce harvest-now risk.

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2	Shor's algorithm on Elliptic-Curve Cryptography (ECC/ECDSA)	ECC relies on the discrete-logarithm problem. Studies cited in a Quantum Zeitgeist review note that a 256-bit ECC key could be broken by a quantum computer with ≈1 million qubits ⁶ . Garn & Kan's 2025 resource estimates indicate that computing discrete logarithms for a 256-bit ECC key might require ~370 000 physical qubits and could run in ~8-12 hours on a fault-tolerant device ⁴ . Because ECC keys are shorter, fewer qubits are needed than for RSA.	8/10 – very high risk: ECC is widely deployed (TLS, Bitcoin) and appears easier to break than RSA. Q-day for ECC could arrive earlier than for RSA.	Adopt PQC and hybrid signatures. Replace ECDH and ECDSA with post-quantum schemes (e.g., lattice-based ML-KEM and ML-DSA). Use hybrid certificates combining ECC with PQC signatures until PQC is widely available. For cryptocurrency, upgrade to quantum-secure signature schemes or implement multi-signature/hybrid wallets.

#	Attack/test (quantum algorithm)	Metrics & complexity	Risk score	Protection/ mitigation
3	Harvest-now/ Decrypt-later attack	Attackers record encrypted traffic now and decrypt once quantum computers mature. Cloudflare warns that “harvest-now/decrypt-later” attacks are a real threat; Q-day might come within ~15 years ⁷. RSA-2048 keys can potentially be broken with <1 million qubits in under a week ², so data harvested today could be decrypted in the 2030s.	9/10 – imminent: Data stolen today could be readable later; there is no need to break encryption now.	Re-encrypt long-term data with PQC. Implement forward secrecy using PQC key-exchange (e.g., ML-KEM) for TLS. Rotate keys frequently and shorten certificate lifetimes. Organisations with sensitive archives should re-encrypt or destroy data that must remain confidential beyond the expected Q-day.
4	Grover’s algorithm for symmetric key search (AES)	Grover reduces brute-force search complexity from 2^n to $2^{\lceil n/2 \rceil}$. For AES-128, Grover requires $\approx 2^{64}$ operations, reducing security to 64 bits ⁸. AES-256 drops to 2^{128} operations, equivalent to 128-bit security, which is still considered strong ⁹. Grover’s algorithm must run sequentially, so parallelising it yields diminishing returns ¹⁰.	5/10 – moderate: Generic symmetric encryption is weakened but not broken. AES-256 remains secure for decades.	Use 256-bit keys. Upgrade systems from AES-128 to AES-256. Consider segmented or multi-party key encryption (e.g., Freemindtronic’s segmented key scheme) to avoid a single key exposure ¹¹. Rotate keys and limit key reuse.

#	Attack/test (quantum algorithm)	Metrics & complexity	Risk score	Protection/ mitigation
5	Grover's algorithm for hash preimage attacks	For an n-bit hash, Grover finds a preimage in $2^{\{n/2\}}$ steps ¹² . SHA-256 drops from 256-bit preimage security to 128-bit , SHA-512 drops to 256-bit ¹³ . Hashes smaller than 256 bits become very weak; for example, SHA-224 reduces to 112-bit security ¹⁴ .	5/10 – moderate: Preimage search becomes easier but still requires astronomical work; poor-quality hashes (≤ 224 bits) are unacceptable.	Use larger hash outputs. Move to SHA-256 or SHA-512 for long-term security ¹³ . Double the hash length if necessary. For password hashing, combine with salt and key-stretching (PBKDF2, Argon2) to counter brute-force improvements.
6	BHT (Brassard-Høyer-Tapp) quantum collision attack on hash functions	BHT reduces collision complexity from $O(2^{\{n/2\}})$ to $O(2^{\{n/3\}})$ ¹⁵ . SHA-256 collision security drops from 128 bits to roughly 85 bits ¹⁶ . However, BHT requires storing $\approx 2^{\{n/3\}}$ intermediate hash outputs; for SHA-256 this means $\sim 2^{\{85\}}$ quantum memory elements ¹⁷ —an impractical requirement.	3/10 – low: The algorithm is theoretically interesting but requires astronomical memory and time; classical parallel methods outperform it ¹⁸ .	Use longer hashes and diversify hash functions. SHA-384 or SHA-512 restore ~ 128 -bit collision security even under BHT ¹⁹ . Avoid truncated hashes and use keyed hashes (HMAC).

#	Attack/test (quantum algorithm)	Metrics & complexity	Risk score	Protection/ mitigation
7	Simon's algorithm on block-cipher structures (Even-Mansour / 3-round Feistel)	Simon's hidden-period algorithm can break certain simplified ciphers. Kuwakado and Morii showed that Even-Mansour and 3-round Feistel constructions can be broken in polynomial time ²⁰ . These attacks exploit specific structures and do not extend to full-round AES or modern ciphers.	2/10 – very low: Limited to toy constructions; not feasible on production systems.	Use well-designed ciphers with many rounds. Modern block ciphers (AES, ChaCha20) avoid simple structures and remain unbroken by Simon-type attacks. Designers should ensure that new modes do not exhibit hidden periodicities.
8	BV-based quantum differential cryptanalysis	Quantum variants of differential cryptanalysis use the Bernstein–Vazirani (BV) algorithm. Researchers show that the initial phase to discover useful differences can run in polynomial time ²¹ , and the key-recovery phase achieves a quadratic speed-up compared with classical differential attacks ²² . Some BV-based attacks can reduce the complexity of classical differential cryptanalysis from $O(n^3q^2 \log n)$ to $O(nq^2)$ ²³ .	4/10 – low-moderate: Applicable to specially constructed ciphers; no practical attack on AES yet.	Increase rounds and non-linearity. Use ciphers that resist differential analysis and adopt cryptographic designs evaluated under quantum models. Maintain large S-boxes and diffusion to hinder these attacks.

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9	Variational Quantum Attack (VQAA) on symmetric ciphers	VQAA uses a variational quantum algorithm to search the key space by encoding the ciphertext as the ground state of a Hamiltonian. The proposal shows that the VQAA uses roughly the same order of search-space queries as Grover's algorithm and sometimes outperforms it ²⁴ . Simulations demonstrate that a gradient-descent-based VQAA can converge faster than Grover's for simplified DES (S-DES) instances ²⁴ . The speedup depends on entanglement in the ansatz and is unproven for large key sizes.	5/10 – moderate: VQAA is an emerging technique; current results are on toy ciphers with few bits, but the possibility of beating Grover for real ciphers is concerning.	Adopt higher key lengths and monitor research. Use AES-256 or ChaCha20 with 256-bit keys, which provide 128-bit post-quantum security even if search algorithms improve modestly. Stay informed about advances in variational quantum attacks and update cryptographic standards accordingly.

#	Attack/test (quantum algorithm)	Metrics & complexity	Risk score	Protection/ mitigation
10	Quantum side-channel and machine-learning attacks	Quantum computers may enhance side-channel attacks by extracting keys from timing, power or electromagnetic leakage. A Quantum Zeitgeist article notes that quantum computers could perform “quantum side-channel attacks” by exploiting information about the implementation of an encryption algorithm rather than the algorithm itself ²⁵. Such attacks could also leverage quantum machine-learning to identify patterns in leaked data ²⁶. There are no concrete qubit/time estimates because the attacks depend on implementation details.	7/10 – high risk (implementation-dependent): Once quantum hardware becomes practical, it could amplify existing side-channel threats.	Harden implementations. Apply constant-time coding, noise injection, and shielding against physical leakage. Use hardware security modules (HSMs) or tamper-resistant chips. Continually monitor side-channel research and update countermeasures for quantum settings. Post-quantum algorithms are not immune to implementation attacks, so secure coding practices remain essential.

Additional recommendations

- 1. Plan for post-quantum migration now.** National regulators and industry (NSA, NIST, UK NCSC) have set migration deadlines between **2030 and 2035** ²⁷. Organisations should inventory cryptographic assets, prioritise high-value data and systems, and develop a migration roadmap to PQC.
- 2. Use hybrid cryptography during transition.** Combining classical and post-quantum algorithms (e.g., X25519 + ML-KEM, ECDSA + ML-DSA) provides defence in depth while standards mature.
- 3. Keep keys and messages short-lived.** Reduce key lifetimes and implement perfect forward secrecy. Short-term session keys limit the utility of harvested ciphertext.
- 4. Monitor quantum-algorithm advances.** New algorithms like Chen's 2024 lattice attack (later shown to be flawed) illustrate that surprises can occur ²⁸. Security teams should watch for breakthroughs in factorization, discrete logarithms or machine-learning-based attacks.

5. **Educate stakeholders.** Many decision-makers underestimate Q-day. Sharing metrics (qubits, runtime) and roadmaps helps justify investments in PQC and secure coding.

Conclusion

Quantum computing is transitioning from theory to practice. While truly large-scale fault-tolerant computers remain years away, research continues to lower the resource requirements for breaking cryptographic schemes. Organisations should not wait for Q-day; by understanding the metrics of quantum attacks and adopting recommended mitigations—doubling symmetric key lengths, migrating to post-quantum algorithms, hardening implementations, and planning for re-encryption—security teams can ensure that data protected today remains safe in the quantum future.

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